Arianna Bunnell

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Education

PhD in Computer Science - ABD 08/23 - 05/28
Department of Information & Computer Sciences, University of Hawai'i at Mānoa

Master's of Science in Computer Science 08/21 - 05/23
Department of Information & Computer Sciences, University of Hawai'i at Mānoa
Thesis Title: Early Breast Cancer Diagnosis via Breast Ultrasound and Deep Learning

Bachelor's of Science in Statistics 08/17 - 05/21
Department of Mathematics & Statistics, Utah State University

Bachelor's of Science in Computer Science 08/17 - 05/21
Department of Computer Science, Utah State University

Research Experience

Graduate Research Assistant 08/21 - *Present*
Perform original research under the supervision of Dr. Peter Sadowski and Dr. John Shepherd in the Information and Computer Sciences Department at the University of Hawai'i at Mānoa.

Undergraduate Research Assistant 04/20 - 07/21
Assisted Dr. John Stevens in the Mathematics and Statistics Department at Utah State University in his research in developing an adjustment to the Stuart-Maxwell test of marginal homogeneity for ordinal classes.

Work Experience

Data Science Intern, Milliman 05/22 - 08/22
Designed and coded a multi-output regression deep neural network for predicting patient cost-based insurance risk scores from their diagnosis and treatment history.

Analytics Engineer Intern, Health Catalyst 04/21 - 07/21
Worked on a project developing a new measure for health inequity for continuous-valued health outcomes in an inpatient facility based on the Gini coefficient. Also coded query auditing software for automated query performance and design analysis.

Mathematics & Statistics Teaching Fellow, Utah State University 01/19 - 05/21
Supported instructor with grading, lesson planning, assessment design, and one-on-one student instruction at Utah State University. Experience in the following courses: Business Statistics, Biostatistics, Design of Experiments, Linear Regression & Time Series, and Introduction to Statistics.

Research and Operations Intern, Principal Financial Group 05/20 - 12/20
Designed an internal tool for the research and deployment of investment products for the Data Science and Research Team at Principal Global Investors.

Data Science Intern, UnitedHealth Group Research & Development 02/19 - 08/19
Led a team of three other interns to design and implement an algorithm in Python to predict hyperglycemic events in Type II diabetics in the Medicare population.

Publications

A. Bunnell and S. Rowe. The Addition of Artificial Intelligence to Consent-as-Authority: Development of Care-Ethical Consent-as-Trust. *preprint.*

A. Bunnell, D. Valdez, F. Strand, Y. Glaser, P. Sadowski and J. A. Shepherd. (2025). Artificial intelligence-enhanced handheld breast ultrasound for screening: A systematic review of diagnostic test accuracy. *PLOS Digital Health*, 4(9), p.e0001019.

A. Bunnell, D. Cataldi, Y. Glaser, T. Wolfgruber, S. Heymsfield, A. B. Zonderman, T. L. Kelly, P. Sadowski, J. A. Shepherd. Deep Learning Enables Large-Scale Shape and Appearance Modeling in Total-Body DXA Imaging. *International Workshop on Shape in Medical Imaging* Cham: Springer Nature Switzerland, 2025.

T. Beckstrom, **A. Bunnell**, T. M. Maaz, M. B. Kantar, J. L. Deenik, C. T. Glazer, P. Sadowski, and S. E. Crow. "Mid-infrared spectroscopy and machine learning improve accessibility of Hawaii soil health assessment." *Soil Science Society of America Journal*, 89, no. 3 (2025): e70081.

A. Bunnell, D. Valdez, T. K. Wolfgruber, B. Quon, K. Hung, B. Y. Hernandez, T. B. Seto, J. Killeen, M. Miyoshi, P. Sadowski and J. A. Shepherd. (2025). Prediction of mammographic breast density based on clinical breast ultrasound images using deep learning: a retrospective analysis. *The Lancet Regional Health - Americas*, (46), p.101096.

A. Bunnell, K. Hung, J. A. Shepherd and P. Sadowski. (2024). BUSClean: Open-source software for breast ultrasound image pre-processing and knowledge extraction for medical AI. *PLOS ONE*, 19(12), p.e0315434.

A. Bunnell, Y. Glaser, D. Valdez, T. Wolfgruber, A. Altamirano, C. Zamora Gonzalez, B. Y. Hernandez, P. Sadowski, J. A. Shepherd. Learning a Clinically-Relevant Concept Bottleneck for Lesion Detection in Breast Ultrasound. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI);* Cham: Springer Nature Switzerland, 2024.

D. Valdez, **A. Bunnell,** S. Lim, P. Sadowski, & J. A. Shepherd. (2024). Performance of progressive generations of GPT on an exam designed for certifying physicians as Certified Clinical Densitometrists. *Journal of Clinical Densitometry*, 27(2), 101480.

N. Z. Zemi*, **A. Bunnell*,** D. Valdez, & J. A. Shepherd, (2024, May). Assessing the feasibility of AI-enhanced portable ultrasound for improved early detection of breast cancer in remote areas. in *17th International Workshop on Breast Imaging (IWBI 2024)* (Vol. 13174, pp. 88-94). SPIE. *Co-first author

A. Bunnell and S. Rowe. The Effect of AI-Enhanced Breast Imaging on the Caring Radiologist-Patient Relationship. in *Pacific Symposium on Biocomputing.* 2023: Kohala Coast, Hawaii, USA, 3–7 January 2023. 2022. World Scientific.

Presentations

A. Bunnell, S. Yang, N. Zemi, D. Valdez, B. Hernandez, P. Sadowski and J. A. Shepherd. Adaptation of a Histopathology Foundation AI Model for Cytopathology Interpretation of Breast Cancer to Improve Diagnostic Access [oral presentation]. *IARC@60 International Scientific Conference.* May 2026.

K. Saito, W. Huynh, E. Saloma **A. Bunnell**, P. Sadowski, and J. A. Shepherd Developing a Data Standardization Pipeline for Evaluation of Mammography AI in Hawaii [poster presentation]. *Undergraduate Research Opportunities Program Symposium.* May 2026.

A. Bunnell, S. Yang, E. Cho, T. Wolfgruber, K. Cassel, J. A. Shepherd, and P. Sadowski. Collaborative Mixing of Clinical Concept Sources for Explainable Malignancy Prediction in Dermoscopy [poster presentation]. *Biomedical Sciences & Health Disparities Symposium.* Apr 2026.

A. Bunnell, D. Cataldi, Y. Glaser, T. Wolfgruber, S. Heymsfield, A. B. Zonderman, T. L. Kelly, P. Sadowski, J. A. Shepherd. Deep Learning Enables Large-Scale Shape and Appearance Modeling in Total-Body DXA Imaging [poster presentation]. *International Workshop on Shape in Medical Imaging* Sept 2025.

A. Bunnell, T. Wolfgruber, B. Quon, C. Kane, D. Valdez, B. Hernandez, K. Kerlikowske, P. Sadowski and J. A. Shepherd. Mammography AI Models and Radiomic Features for Breast Cancer Risk Prediction: A Matched Case-Control Study in an Ethnically-Diverse Cohort [oral presentation]. *11th International Breast Density & Cancer Risk Assessment Workshop.* Jun 2025.

A. Bunnell, Y. Glaser, D. Valdez, T. Wolfgruber, A. Altamirano, C. Zamora Gonzalez, B. Y. Hernandez, P. Sadowski, J. A. Shepherd. Learning a Clinically-Relevant Concept Bottleneck for Lesion Detection in Breast Ultrasound [oral presentation]. *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI).* Oct 2024.

C. Kane, **A. Bunnell**, P. Sadowski and J. A. Shepherd. Customization of AI-Based Breast Cancer Risk Models for Validation in AANHPI Women [oral presentation]. *Summer Undergraduate Research Experience Symposium.* Aug 2024.

A. Bunnell, D. Cataldi, Y. Glaser, T. Wolfgruber, P. Sadowski and J. A. Shepherd. Fiducial Point Placement on Total-Body DXA Scans with Deep Learning [oral presentation]. *Biomedical Sciences & Health Disparities Symposium.* Apr 2024.

A. Bunnell, D. Valdez, F. Strand, Y. Glaser, P. Sadowski, & J. A. Shepherd. Is AI-enhanced breast ultrasound ready for breast cancer screening in low-resource environments? A systematic review [poster presentation]. *AACR Annual Meeting.* Apr 2024.

K. Hung, **A. Bunnell**, P. Sadowski and J. A. Shepherd. Annotation Parsing in Clinical, Real-World Breast Ultrasound Imaging Data [poster presentation]. *Summer Undergraduate Research Experience Symposium.* Aug 2023.

A. Bunnell, D. Valdez, T. Wolfgruber, B. Quon, L. Leong, J. Fukui, B. Hernandez, Y. Shvetsov, P. Washington, P. Sadowski and J. A. Shepherd. Artificial Intelligence Predicts Mammographic Breast Density from Clinical Breast Ultrasound Images [oral presentation]. *10th International Breast Density & Cancer Risk Assessment Workshop.* Jun 2023.

L. Leong, T. Wolfgruber, B. Quon, **A. Bunnell**, J. Fukui, B. Hernandez, Y. Shvetsov, K. Kerlikowske and J. A. Shepherd. Image-Based Models for Predicting Advanced Breast Cancer Risk [oral presentation]. *10th International Breast Density & Cancer Risk Assessment Workshop.* Jun 2023.

A. Bunnell, D. Valdez, T. Wolfgruber, J. Fukui, P. Sadowski and J. A. Shepherd. Data Standardization of Clinical, Real-World Breast Ultrasound Imaging Data [poster presentation]. *Biomedical Sciences & Health Disparities Symposium.* Apr 2023.

A. Bunnell and S. Rowe. The Effect of AI-Enhanced Breast Imaging on the Caring Radiologist-Patient Relationship [oral presentation]. *Pacific Symposium on Biocomputing.* Jan 2023.

A. Bunnell, D. Valdez, T. Wolfgruber, B. Hernandez, P. Sadowski and J. A. Shepherd. Artificial Intelligence Detects, Classifies, and Describes Lesions in Clinical Breast Ultrasound Images [poster presentation]. *San Antonio Breast Cancer Symposium.* Dec 7, 2022.

J. Stevens and **A. Bunnell**. Extending the Stuart-Maxwell Test to Account for Ordinal Category Levels [oral presentation]. *NCCC-170 Research Advances in Agricultural Statistics Annual Meeting.* June 25, 2021.

Professional Activities

AIPHI Affinity Group Scientific Program Director 05/25 - *Present*
Direct scientific programming for a recurring seminar series focused on artificial intelligence in cancer research, including identification and recruitment of expert speakers across computer science, engineering, radiology, and clinical oncology.

Conference Reviewing
International Conference on Medical Image Computing and Computer Assisted Intervention

Journal Reviewing
Journal of Computational Design and Engineering; SoftwareX; PLOS Digital Health; Journal of Medical Imaging; Quantitative Imaging in Medicine and Surgery; PLOS One.

Undergraduate Mentorship 05/23 - *Present*
I have had the opportunity to mentor several undergraduate students in computer science in the completion of independent research projects, typically over the summer semester. Past mentees: Samuel Yang (May 2025 - May 2026) MS in CS in May 2026; Kailee Hung (May 2023 - May 2024) BS in CS in May 2024; Cade Kane (May 2024 - August 2024) BS in CS in May 2025; Elijah Saloma (August 2025 - May 2026) BS in CS in May 2026; Wilson Huynh (August 2025 - May 2026) BS in CS in May 2026; and Kanta Saito (August 2025 - May 2026) BS in CS in May 2027.

Program Committees
Women in Machine Learning Social Chair at ICML 2023.
Women in Computer Vision Social Chair at ICCV 2025.

Leadership & Service Activities

GWISH President 05/24 - *Present*
Design and host professional development events for the Graduate Women in Science Hawaii (GWISH) organization, as well as manage all official communications and activities of the organization. Wrote grants resulting in over $12,000 in funding awarded to the organization.

Graduate Student Community Engagement Co-Chair 08/23 - 06/26
My co-chair and I regularly apply for university grants (more than $5,500 awarded to date) to organize weekly Coffee Hours for graduate students and faculty to foster community within the Information and Computer Science Department at UH Mānoa.

Sci-MI Computer Science Program Director 04/23 - 11/24

Managed a team of ≈25 undergraduate and graduate instructors in creating content for a summer program that mentors underrepresented high school students. Mentor high school students one-on-one in computational neuroscience- and computer science-focused research projects with the goal of getting them prepared for careers in science and college applications. Also composed and presented lecture content related to deep learning and data visualization in the sciences.

GWISH Professional Development Coordinator 05/23 - 05/24

Design and host professional development events targeted at female graduate students in STEM for the Graduate Women in Science Hawaii (GWISH) organization.

Fellowships & Awards

Soroptimist Founder Region Fellow 05/26

NIH AIM AHEAD FHIR Trainee 02/26 - 07/26

Peterson Award at UH Mānoa 05/25

Sarah Ann Martin ARCS Award in Information and Computer Science 05/24

Google Computer Science Research Mentorship Program Fellow 08/23 - 12/23

Hawai'i Data Science Institute Fellow 08/23 - 07/24

NIH AIM AHEAD 2023 Health Equity Data Challenge 3rd Place Winner 08/23

NIH AIM AHEAD HEART Fellow/Graduate Mentor 11/22 - 05/23